RIPHAH INTERNATIONAL UNIVERSITY



Artificial Intelligence

LAB #9

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Date of Submission

5/11/2026

Task 1:

Here are the step-by-step actions performed by the agent for three different initial room conditions

Combination 1: {"A": "Clean", "B": "Dirty"} **(Starting at Location "A")**

- **Step 1:** Location: A | Room Status: Clean | Action: Move Right | Environment: {'A': 'Clean', 'B': 'Dirty'}
- **Step 2:** Location: B | Room Status: Dirty | Action: Suck | Environment: {'A': 'Clean', 'B': 'Clean'}
- **Step 3:** Location: B | Room Status: Clean | Action: Move Left | Environment: {'A': 'Clean', 'B': 'Clean'}

Combination 2: {"A": "Clean", "B": "Clean"} **(Starting at Location "A")**

- **Step 1:** Location: A | Room Status: Clean | Action: Move Right | Environment: {'A': 'Clean', 'B': 'Clean'}
- **Step 2:** Location: B | Room Status: Clean | Action: Move Left | Environment: {'A': 'Clean', 'B': 'Clean'}
- **Step 3:** Location: A | Room Status: Clean | Action: Move Right | Environment: {'A': 'Clean', 'B': 'Clean'}

Combination 3: {"A": "Dirty", "B": "Clean"} **(Starting at Location "A")**

- **Step 1:** Location: A | Room Status: Dirty | Action: Suck | Environment: {'A': 'Clean', 'B': 'Clean'}
- **Step 2:** Location: A | Room Status: Clean | Action: Move Right | Environment: {'A': 'Clean', 'B': 'Clean'}
- **Step 3:** Location: B | Room Status: Clean | Action: Move Left | Environment: {'A': 'Clean', 'B': 'Clean'}